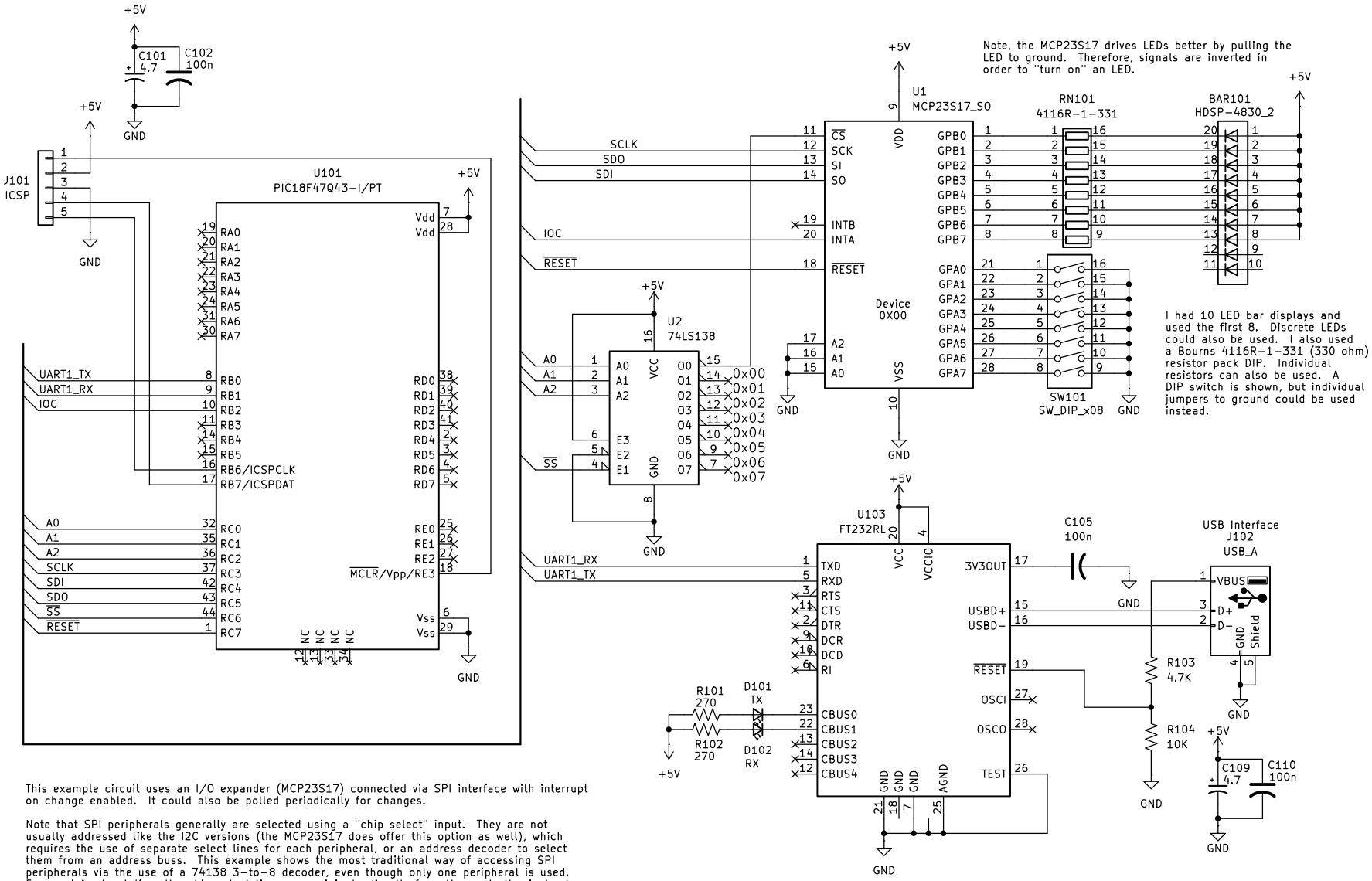

This example circuit uses an I/O expander (MCP23S17) connected via SPI interface with interrupt on change enabled. It could also be polled periodically for changes.

Note that SPI peripherals generally are selected using a "chip select" input. They are not usually addressed like the I2C versions (the MCP23S17 does offer this option as well), which requires the use of separate select lines for each peripheral, or an address decoder to select them from an address buss. This example shows the most traditional way of accessing SPI peripherals via the use of a 74138 3-to-8 decoder, even though only one peripheral is used. For a minimal solution, the chip select lines can originate directly from the controller instead.